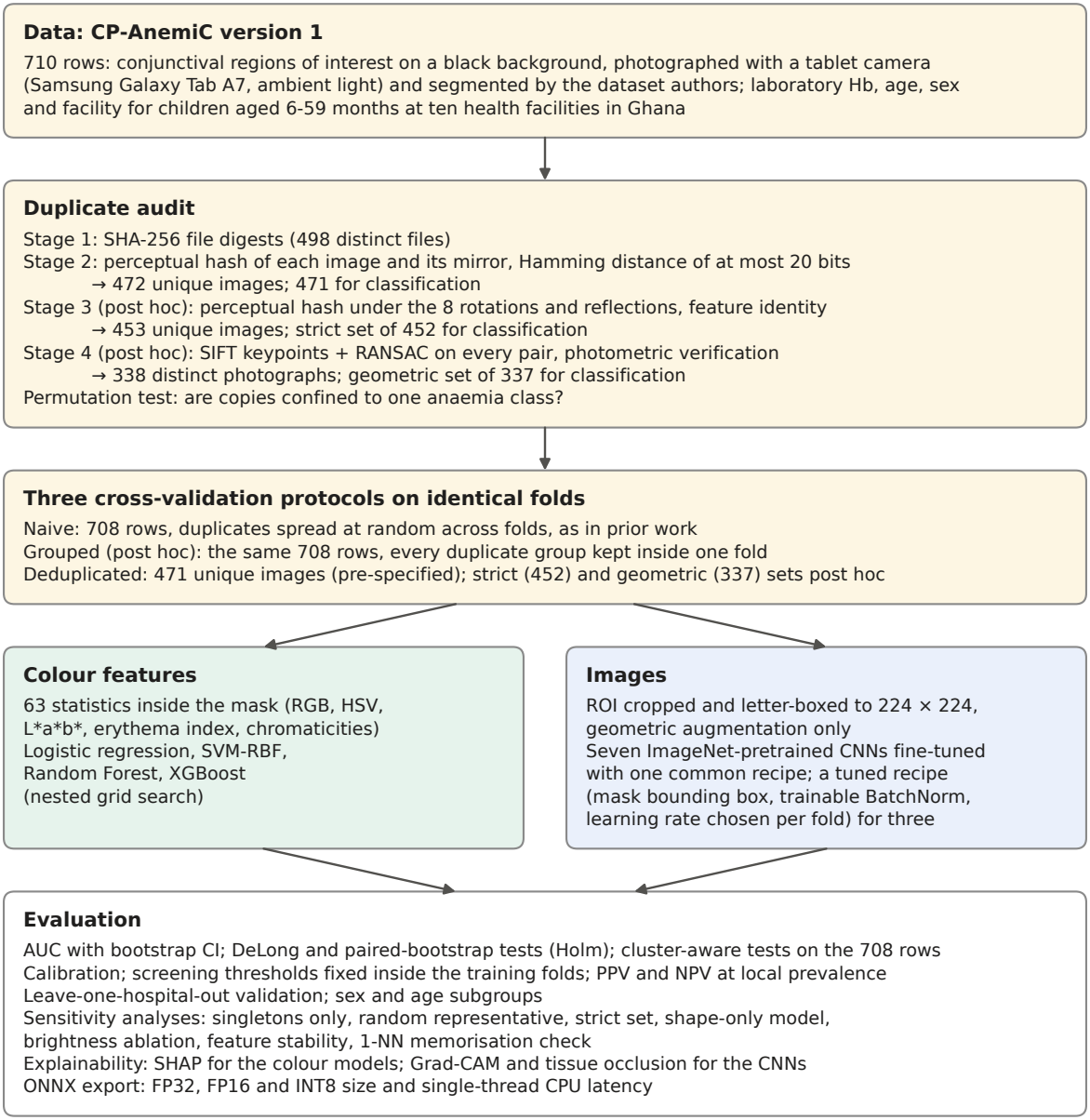
Data: CP-AnemiC version 1
710 rows: conjunctival regions of interest on a black background, photographed with a tablet camera (Samsung Galaxy Tab A7, ambient light) and segmented by the dataset authors; laboratory Hb, age, sex and facility for children aged 6-59 months at ten health facilities in Ghana
Duplicate audit
Stage 1: SHA-256 file digests (498 distinct files)
Stage 2: perceptual hash of each image and its mirror, Hamming distance of at most 20 bits
→ 472 unique images; 471 for classification
Stage 3 (post hoc): perceptual hash under the 8 rotations and reflections, feature identity
→ 453 unique images; strict set of 452 for classification
Stage 4 (post hoc): SIFT keypoints + RANSAC on every pair, photometric verification
→ 338 distinct photographs; geometric set of 337 for classification
Permutation test: are copies confined to one anaemia class?
Three cross-validation protocols on identical folds
Naive: 708 rows, duplicates spread at random across folds, as in prior work
Grouped (post hoc): the same 708 rows, every duplicate group kept inside one fold
Deduplicated: 471 unique images (pre-specified); strict (452) and geometric (337) sets post hoc
Colour features
63 statistics inside the mask (RGB, HSV, L*a*b*, erythema index, chromaticities)
Logistic regression, SVM-RBF, Random Forest, XGBoost
(nested grid search)
Images
ROI cropped and letter-boxed to 224 × 224, geometric augmentation only
Seven ImageNet-pretrained CNNs fine-tuned with one common recipe; a tuned recipe (mask bounding box, trainable BatchNorm, learning rate chosen per fold) for three
Evaluation
AUC with bootstrap CI; DeLong and paired-bootstrap tests (Holm); cluster-aware tests on the 708 rows
Calibration; screening thresholds fixed inside the training folds; PPV and NPV at local prevalence
Leave-one-hospital-out validation; sex and age subgroups
Sensitivity analyses: singletons only, random representative, strict set, shape-only model, brightness ablation, feature stability, 1-NN memorisation check
Explainability: SHAP for the colour models; Grad-CAM and tissue occlusion for the CNNs
ONNX export: FP32, FP16 and INT8 size and single-thread CPU latency